In any triangle ABC, prove that (Q1 to Q16)

1. $\sin\frac{B-C}{2}=\frac{b-c}{a}\cos\frac{A}{2}$

Ans.

Sol. $\text{RHS}=\left(\frac{b-c}{a}\right)\cos\frac{A}{2}=\left(\frac{2R\sin B-2R\sin C}{2R\sin A}\right)\cos\frac{A}{2}=\frac{2\cos\left(\frac{B+C}{2}\right)\cdot\sin\left(\frac{B-C}{2}\right)}{2\sin\frac{A}{2}\cos\frac{A}{2}}\times\cos\frac{A}{2}$

$$=\frac{\sin\frac{A}{2}\cdot\sin\left(\frac{B-C}{2}\right)}{\sin\frac{A}{2}}=\sin\left(\frac{B-C}{2}\right)=\text{LHS}$$

2. $a(b\cos C-c\cos B)=b^2-c^2$

Ans.

Sol. $\text{LHS}=a(b\cos C-c\cos B)=(b\cos C+c\cos B)(b\cos C-c\cos B)=b^2\cos^2C-c^2\cos^2B$

$$=a\times\left[b\times\left(\frac{a^2+b^2-c^2}{2ab}\right)-c\times\left(\frac{c^2+a^2-b^2}{2ca}\right)\right]=\frac{2\left(b^2-c^2\right)}{2}=b^2-c^2$$

3. $a(\cos B+\cos C)=2(b+c)\sin^2\frac{A}{2}$

Ans.

Sol. $\text{LHS}=a(\cos B+\cos C)=a\left(\frac{c^2+a^2-b^2}{2ca}+\frac{a^2+b^2-c^2}{2ab}\right)=\frac{1}{2}\left[c+\frac{a^2}{c}-\frac{b^2}{c}+\frac{a^2}{b}\right]$

$$=\left(\frac{bc^2+a^2b-b^3+a^2c+b^2c-c^3}{2bc}\right)=\left[\frac{b\left(c^2-b^2\right)+c\left(b^2-c^2\right)+a^2\left(b+c\right)}{2bc}\right]$$

$$=\frac{a^2\left(b+c\right)-\left(b^2-c^2\right)\left(b-c\right)}{2bc}=\frac{\left(b+c\right)}{2bc}\left[a^2-\left(b-c\right)^2\right]=\left(\frac{b+c}{2bc}\right)\left(a+c-b\right)\left(a+b-c\right)$$

$$=\left(\frac{b+c}{2bc}\right)\times 2(s-b)\times 2(s-c)=2(b+c)\times\frac{(s-b)(s-c)}{bc}=2(b+c)\sin^2\frac{A}{2}=\text{RHS}$$

4. $\dfrac{a+b}{a-b}=\tan\dfrac{A+B}{2}\cot\dfrac{A-B}{2}$

Ans.

Sol. $\text{LHS}=\dfrac{a+b}{a-b}=\dfrac{2R(\sin A+\sin B)}{2R(\sin A-\sin B)}=\dfrac{2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)}{2\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)}=\tan\left(\frac{A+B}{2}\right)\cdot\cot\left(\frac{A-B}{2}\right)=\text{RHS}$

5. $a\sin\left(\dfrac{A}{2}+B\right)=(b+c)\sin\dfrac{A}{2}$

Ans.

Sol. $\text{LHS}=a\sin\left(\dfrac{A}{2}+B\right)=a\sin\dfrac{A}{2}\cos B+a\cos\dfrac{A}{2}\sin B=2R\sin A\sin\dfrac{A}{2}\cos B+2R\sin A\cos\dfrac{A}{2}\sin B$

$$=(2R)\sin\frac{A}{2}\left[2\sin\frac{A}{2}\cos\frac{A}{2}\cos B+2\cos\frac{A}{2}\cos\frac{A}{2}\sin B\right]\times 2$$

$$\text{LHS}=a\sin\left(\frac{A+B}{2}+\frac{B}{2}\right)=a\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{B}{2}\right)+a\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{B}{2}\right)$$

$$=a\cos\frac{C}{2}\cos\frac{B}{2}+a\sin\frac{C}{2}\sin\frac{B}{2}=a\cos\left(\frac{B-C}{2}\right)$$

$$=2R\left(2\sin\frac{A}{2}\cos\frac{A}{2}\cos\frac{C}{2}\cos\frac{B}{2}+2\sin\frac{A}{2}\cos\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right)$$

$$\text{OR simultaneously}\Rightarrow a\sin\left(\frac{A}{2}+B\right)=(b+c)\sin\frac{A}{2}$$

$$\Rightarrow\frac{b+c}{a}=\frac{\sin\left(\frac{A}{2}+B\right)}{\sin\frac{A}{2}}$$

$$\Rightarrow\frac{\sin B+\sin C}{\sin A}=\frac{\cos\left(\frac{B-C}{2}\right)}{\sin\frac{A}{2}}$$

$$\Rightarrow \frac{2\sin\left(\frac{B+C}{2}\right)\cos\left(\frac{B-C}{2}\right)}{2\sin\frac{A}{2}\cos\frac{A}{2}} = \frac{\cos\left(\frac{B-C}{2}\right)}{\sin\frac{A}{2}}$$

$$\Rightarrow \frac{\cos\frac{A}{2}}{\cos\frac{A}{2}} = 1 \Rightarrow 1 = 1 \Rightarrow \text{LHS} = \text{RHS}$$

6. $(b+c-a)\left(\cot\frac{B}{2}+\cot\frac{C}{2}\right) = 2a\cot\frac{A}{2}$

Ans.

Sol. $\text{LHS} = (b+c-a)\left(\cot\frac{B}{2}+\cot\frac{C}{2}\right) = 2(s-a)\left[\frac{\cos\frac{B}{2}\sin\frac{C}{2}+\cos\frac{C}{2}\sin\frac{B}{2}}{\sin\frac{B}{2}\sin\frac{C}{2}}\right]$

$$= 2(s-a)\times\frac{\sin\left(\frac{B+C}{2}\right)}{\sqrt{\frac{(s-a)(s-c)}{ca}}\times\sqrt{\frac{(s-a)(s-b)}{ab}}} = 2a\times\frac{\cos\frac{A}{2}}{\sqrt{\frac{(s-b)(s-c)}{bc}}} = 2a\frac{\cos\frac{A}{2}}{\sin\frac{A}{2}}$$

$$= 2a\cot\frac{A}{2} = \text{RHS}$$

7. $(a^2 - b^2 + c^2)\tan B = (a^2 + b^2 - c^2)\tan C$

Ans.

Sol. $\text{LHS} = (a^2 - b^2 + c^2)\tan B = 2ca\left(\frac{a^2+c^2-b^2}{2ac}\right)\tan B = 2ca\times\cos B\times\frac{\sin B}{\cos B}$

$= 2ca\sin B = 8R^2\sin A\sin B\sin C$

$= \text{RHS} = (a^2 + b^2 - c^2)\tan C = 2ab\sin C = 8R^2\sin A\sin B\sin C$

8. $a\sin(B-C) + b\sin(C-A) + c\sin(A-B) = 0$

Ans.

Sol. $\text{LHS} = a\sin(B-C) + b\sin(C-A) + c\sin(A-B)$

$= 2R\sin(B+C)\cdot\sin(B-C) + 2r\sin(C+A)\sin(C-A) + 2R\sin(A+B)\sin(A-B)$

$= 2R[\sin^2B - \sin^2C + \sin^2C - \sin^2A + \sin^2A - \sin^2B] = 0 = \text{RHS}$

9. $a\sin\frac{A}{2}\sin\frac{B-C}{2}+b\sin\frac{B}{2}\sin\frac{C-A}{2}+c\sin\frac{C}{2}\sin\frac{A-B}{2}=0$

Ans.

Sol. $\text{LHS}=2R\times 2\sin\left(\frac{B+C}{2}\right)\cos^2\left(\frac{B+C}{2}\right)\cdot\sin\left(\frac{B-C}{2}\right)+4R\sin\left(\frac{C+A}{2}\right)\sin\left(\frac{C-A}{2}\right)\cdot\sin^2\frac{B}{2}$

$$+4R\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)\sin^2\frac{C}{2}$$

$$=4R\left\{\left[\frac{(s-b)(s-c)}{bc}\right]\left(\sin^2\frac{B}{2}-\sin^2\frac{C}{2}\right)+\left[\frac{(s-c)(s-a)}{ca}\right]\left(\sin^2\frac{C}{2}-\sin^2\frac{A}{2}\right)\right.$$

$$\left.+\left[\frac{(s-a)(s-b)}{ab}\right]\left(\sin^2\frac{A}{2}-\sin^2\frac{B}{C}\right)\right\}$$

$$\text{or LHS}=a\cos\left(\frac{B+C}{2}\right)\sin\left(\frac{B-C}{2}\right)+b\cos\left(\frac{A+C}{2}\right)\sin\left(\frac{C-A}{2}\right)+c\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$$

$$=\frac{a}{2}[\sin B-\sin C]+\frac{b}{2}[\sin C-\sin A]+\frac{c}{2}[\sin A-\sin B]$$

$$=R$$

10. $\frac{b^2-c^2}{a^2}\sin 2A+\frac{c^2-a^2}{b^2}\sin 2B+\frac{a^2-b^2}{c^2}\sin 2C=0$

Ans.

Sol. $\text{LHS}=\frac{4R^2\left(\sin^2 B-\sin^2 C\right)}{4R^2\sin^2 A}\times 2\sin A\cos A+2\left(\frac{\sin^2 C-\sin^2 A}{\sin B}\right)\cos B+2\left(\frac{\sin^2 A-\sin^2 B}{\sin C}\right)\cos C$

$$\text{or }\frac{b^2-c^2}{a^2}\sin 2A=\left(\frac{\sin^2 B-\sin^2 C}{\sin^2 A}\right)\times 2\sin A\cos A=\frac{\sin(B+C)\cdot\sin(B-C)2\cos A}{\sin A}$$

$$=\frac{2\sin(B+C)\cdot\sin(B-C)}{\sin A}\cos A=-2\cos(B+C)\sin(B-C)$$

$$=-(\sin 2B-\sin 2C)=\sin 2C-\sin 2B$$

11. $b^2\sin 2C+c^2\sin 2B=2bc\sin A$

Ans.

Sol. $\text{LHS}=b^2\sin 2C+c^2\sin 2B=4R^2\sin^2 B\sin 2C+4R^2\sin^2 C\sin 2B$

$=4R^2\sin B\sin C(2\sin B\cos C+2\cos B\sin C)$

$=2bc\sin(B+C)=2bc\sin A=\text{RHS}$

12. $(b + c) \cos A + (c + a) \cos B + (a + b) \cos C = a + b + c$

Ans.

Sol. $LHS = (b + c) \cos A + (c + a) \cos B + (a + b) \cos C$

$= 2R [\sin B \cos A + \sin C \cos A + \sin C \cos B + \sin A \cos B + \sin A \cos C + \sin B \cos C]$

$= 2R [\sin (A + BB) + \sin (C + A) + \sin (B + C)] = 2R (\sin A + \sin B + \sin C) = a + b + c = RHS$

13. $\dfrac{\sin(B-C)}{\sin(B+C)} = \dfrac{b^2 - c^2}{a^2}$

Ans.

Sol. $LHS = \dfrac{\sin(B-C)}{\sin(B+C)} = \dfrac{\sin^2 B - \sin^2 C}{\sin^2(B+C)} = \dfrac{\sin^2 B - \sin^2 C}{\sin^2 A} = \dfrac{b^2 - c^2}{a^2} = RHS$

14. $\dfrac{a^2 \sin(B-C)}{\sin B + \sin C} + \dfrac{b^2 \sin(C-A)}{\sin C + \sin A} + \dfrac{c^2 \sin(A-B)}{\sin A + \sin B} = 0$

Ans.

Sol. $\dfrac{a^2 \sin(B-C)}{\sin B + \sin C} = a \times 2R \dfrac{\sin(B+C)\sin(B-C)}{\sin B + \sin C} = 2aR\left(\dfrac{\sin^2 B - \sin^2 C}{\sin B + \sin C}\right) = 2aR(\sin B - \sin C)$

$\Rightarrow LHS = 2R^2(\sin A \sin B - \sin A \sin C + \sin B \sin C - \sin B \sin A + \sin C \sin A - \sin C \sin B)$

$= 0 = RHS$

15. $a^2 + b^2 + c^2 = 2 (bc \cos A + ca \cos B + ab \cos C)$

Ans.

Sol. $RHS = 2 (bc \cos A + ca \cos B + ab \cos C) = 2\left[\dfrac{b^2 + c^2 - a^2}{2} + \dfrac{c^2 + a^2 - b^2}{2} + \dfrac{a^2 + b^2 - c^2}{2}\right]$

$= a^2 + b^2 + c^2 = LHS$

16. $a^3 \cos (B - C) + b^3 \cos (C - A) + c^3 \cos (A - B) = 3abc$

Ans.

Sol. $Ra^2 \times 2\sin(B+C)\cos(B-C) = Ra^2[\sin 2B + \sin 2C]$

$= Ra^2[2\sin B \cos B + 2\sin C \cos C]$

$= a^2 b \cos B + a^2 c \cos C$

$\Rightarrow LHS = a^2b \cos B + a^2c \cos C + b^2c \cos C + b^2a \cos A + c^2a \cos A + c^2b \cos B$

$= ab [a \cos B + b \cos A] + ac [a \cos C + c \cos A] + bc [b \cos C + c \cos B]$

$= 3abc = RHS$

17. In a triangle whose sides are 3, 4 and $\sqrt{37}$ metres respectively, prove that the largest angle is 120°.

त्रिभुज जिसकी भुजाऐं 3, 4 तथा $\sqrt{37}$ हैं, तो सिद्ध किजिए कि इसका सबसे बड़ा कोण 120° होगा।

Ans.

Sol. $a = 3, b = 4, c = \sqrt{37}$

$$\cos C = \frac{9+16-37}{2\times3\times4} = \frac{-12}{2\times12} = -\frac{1}{2}$$

$\Rightarrow C = 120^\circ$

18. The sides of a right-angled triangle are 21 and 28 cm; find the length of the perpendiculars drawn to the hypothenuse from the right angle.

यदि किसी समकोण त्रिभुज कि दो छोटी भुजाऐं 21 तथा 28 cm हैं, तो समकोण वाले शीर्ष से कर्ण पर डाले गये लम्ब की लम्बाई क्या होगी।

Ans. 16.8 cm

Sol. $$\Rightarrow \tan\theta = \frac{21}{28} = \frac{h}{y} \Rightarrow h = \frac{3y}{4} \Rightarrow y = \frac{4h}{3}$$

$$\tan\alpha = \frac{28}{21} = \frac{h}{x} \Rightarrow x = \frac{3h}{4}$$

$$\Rightarrow x + y = 35 \quad \Rightarrow \frac{4h}{3} + \frac{3h}{4} = 35 \qquad \Rightarrow \frac{25h}{12} = 35 \Rightarrow h = \frac{84}{5} = 16.8m$$

19. If in any triangle the ratio of angles to one another as 1 : 2 : 3, prove that the corresponding sides are as $1 : \sqrt{3} : 2$.

यदि किसी त्रिभुज के कोणों का अनुपात 1 : 2 : 3 हैं, तो सिद्ध कीजिए कि इनकी संगत भुजाओं का अनुपात $1 : \sqrt{3} : 2$ है।

Ans.

Sol. $A : B : C = 1 : 2 : 3 \Rightarrow A = 30^\circ, B = 60^\circ, C = 90^\circ$

$$\Rightarrow a : b : c = 2R\sin A : 2R\sin B : 2R\sin C = \frac{1}{2} : \frac{\sqrt{3}}{2} : 1 = 1 : \sqrt{3} : 2$$

20. In any triangle, if $\tan\frac{A}{2} = \frac{5}{6}$ and $\tan\frac{B}{2} = \frac{20}{37}$ find $\tan\frac{C}{2}$, and prove that in this triangle $a + c = 2b$.

यदि किसी त्रिभुज में यदि $\tan\frac{A}{2} = \frac{5}{6}$ तथा $\tan\frac{B}{2} = \frac{20}{37}$ तो $\tan\frac{C}{2}$ का मान ज्ञात करो तथा सिद्ध करें $a + c = 2b$.

Ans. 2/5

Sol. $$\tan\frac{C}{2} = \cot\left(\frac{A+B}{2}\right) = \frac{1}{\tan\left(\frac{A+B}{2}\right)} = \frac{1-\tan\frac{A}{2}\cdot\tan\frac{B}{2}}{\tan\frac{A}{2}+\tan\frac{B}{2}} = \frac{1-\frac{5}{6}\times\frac{20}{37}}{\frac{5}{6}+\frac{20}{37}}$$

$$= \frac{222-100}{185+120} = \frac{122}{305} = \frac{2}{5}$$

$$\Rightarrow a + c = 2b \Rightarrow \sin A + \sin C = 2 \sin B \Rightarrow 2\cos\frac{B}{2}\cos\left(\frac{A-C}{2}\right) = 4\sin\frac{B}{2}\cos\frac{B}{2}$$

$$\Rightarrow \cos\left(\frac{A-C}{2}\right) = 2\cos\left(\frac{A+C}{2}\right) \Rightarrow \cos\frac{A}{2}\cos\frac{C}{2} + \sin\frac{A}{2}\sin\frac{C}{2} = 2\cos\frac{A}{2}\cos\frac{C}{2} - 2\sin\frac{A}{2}\sin\frac{C}{2}$$

$$\Rightarrow \cos\frac{A}{2}\cos\frac{C}{2} = 3\sin\frac{A}{2}\sin\frac{C}{2} \Rightarrow 3\tan\frac{A}{2}\tan\frac{C}{2} = 1$$

$$\Rightarrow 3\times\frac{5}{6}\times\frac{2}{5} = 1 \qquad \Rightarrow 1 = 1 \qquad \Rightarrow \text{L.H.S.} = \text{R.H.S.}$$

21. In an isosceles right-angled triangle a straight line is drawn from the middle point of one of the equal sides to the opposite angle. Show that it divides the angle into parts whose cotangents are 2 and 3.
यदि किसी समद्विबाहु समकोण त्रिभुज कि दो समान भुजाओं में से एक भुजा के मध्य बिन्दु को सामने वाले कोण के शीर्ष से मिलाया गया। सिद्ध कीजिए कि यह कोण को दो भागों में बाँटता है जिनके कोस्पर्शज्या के मान 2 तथा 3 है।

Ans. $\alpha + \beta = 45^\circ$

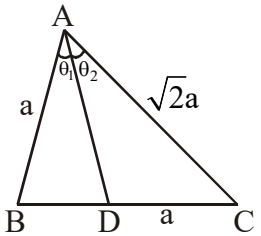
$$\Rightarrow \cot\alpha = \frac{x}{\frac{x}{2}} = 2$$

$$\tan(\alpha+\beta) = 1 \Rightarrow \frac{\frac{1}{2}+\tan\beta}{1-\frac{1}{2}\tan\beta} = 1$$

$$\Rightarrow \frac{1}{2} + \tan\beta = 1 - \frac{1}{2}\tan\beta \Rightarrow \frac{3}{2}\tan\beta = \frac{1}{2}$$

$$\Rightarrow \cot\beta = 3$$

Sol. In ΔABD



$$\tan\theta_1 = \frac{\frac{a}{2}}{a} = \frac{1}{2}$$

$$\cot\theta_1 = 2$$

22. The perpendicular AD to the base of a triangle ABC divides it into segments such that BD, CD and AD are in the ratio of 2, 3 and 6; prove that the vertical angle of the triangle is 45°.
त्रिभुज ABC में शीर्ष लम्ब AD सामने वाली भुजा को इस प्रकार से बाँटता है कि भुजाऐं BD, CD तथा AD क्रमशः 2, 3, 6 के अनुपात में है, तो सिद्ध कीजिए कोण A 45° होगा।

Ans.

Sol. BD : CD : AD = 2 : 3 : 6

$$\tan\alpha = \frac{1}{3} \quad ; \quad \tan\beta = \frac{1}{2}$$

$$\Rightarrow \tan(\alpha+\beta) = \frac{\frac{1}{3}+\frac{1}{2}}{1-\frac{1}{3}\cdot\frac{1}{2}} = \frac{\frac{5}{6}}{\frac{5}{6}} = 1 \Rightarrow \alpha+\beta = 45^\circ$$

23. A ring, 10 cm in diameter, is suspended from a point 12 cm above its centre by 6 equal strings attached to its circumference at equal intervals. Find the cosine of the angle between consecutive strings.
यदि एक 10 cm व्यास वाले चक्र को 12 cm ऊँचाई वाले किसी बिन्दु से चक्र की परिधि पर समान अन्तराल पर छः स्थानों पर एक पतले धागे से बांधा गया, तो लगातार धागों के बीच में कोण की कोज्या का मान ज्ञात करों।

Ans. $\frac{313}{338}$

Sol. $\cos\alpha = \frac{169\times2-25}{2\times13\times13} = \frac{313}{338}$

24. If a^2, b^2 and c^2 be in A.P. prove that cot A, cot B and cot C are in A.P. also.
यदि a^2, b^2 तथा c^2 A.P में है, तो सिद्ध कीजिए cot A, cot B तथा cot C भी A.P. में होगे।

Ans.

Sol. cot A, cot B, cot C $\rightarrow$ AP

$$\Rightarrow \frac{b^2+c^2-a^2}{h\Delta} = \frac{c^2+a^2-b^2}{h\Delta} = \frac{a^2+b^2-c^2}{h\Delta} \rightarrow \text{AP}$$

$\Rightarrow b^2 + c^2 - a^2, c^2 + a^2 - b^2, a^2 + b^2 - c^2 \rightarrow$ AP

$\Rightarrow -2a^2, -2b^2, -2c^2 \rightarrow$ AP

$\Rightarrow a^2, b^2, c^2 \rightarrow$ AP

1. If in a ΔABC, $\Delta = a^2 - (b-c)^2$, then $\tan A =$
यदि ΔABC, $\Delta = a^2 - (b-c)^2$, तो $\tan A =$
(A) 15/16 (B)8/15 (C) 8/17 (D) 1/2

Ans. **B**

Sol. $\Delta = a^2 - (b-c)^2$

$$\tan A = \frac{\sin A}{\cos A} = \frac{2bc\sin A}{b^2+c^2-a^2} = \frac{4\Delta}{b^2+c^2-a^2} = \frac{4\left[a^2-b^2-c^2+2bc\right]}{b^2+c^2-a^2} = \frac{4\times(a-b+c)(a+b-c)}{(b+c+a)(b+c-a)}$$

$$\Rightarrow \tan A = \frac{4\times 2(s-b)\times 2(s-c)}{2s\times 2(s-a)} = 4\times\tan^2\frac{A}{2}$$

$$\Rightarrow \frac{2\tan\frac{A}{2}}{1-\tan^2\frac{A}{2}} = 4\tan^2\frac{A}{2}$$

$$\Rightarrow 1 = 2\tan\frac{A}{2} - 2\tan^3\frac{A}{2}$$

OR $\Rightarrow \Delta = a^2 - b^2 - c^2 + 2bc \Rightarrow b^2 + c^2 - a^2 = 2bc - \Delta$

$$\Rightarrow \frac{b^2+c^2-a^2}{\Delta} = \frac{2bc}{\Delta} - 1$$

$$\Rightarrow 4\cot A = \frac{2bc}{\frac{1}{2}bc\sin A} - 1 \qquad \Rightarrow 4\cot A = 4\,\text{cosec}\,A - 1$$

$$\Rightarrow -\left(\text{cosec}A - \cot A = \frac{1}{4}\right)$$

$$\Rightarrow \text{cosec}A + \cot A = 4$$

$$2\cot A = 4 - \frac{1}{4} = \frac{15}{4}$$

$$\Rightarrow \tan A = \frac{8}{15}(B)$$

2. In a triangle, the lengths of the two larger sides are 10 and 9. If the angles are in AP, then the length of the thrid side can be
यदि किसी त्रिभुज की दो बड़ी भुजाऐं 10 तथा 9 है, तथा कोण AP में है, तो इसकी तीसरी भुजा हो सकती है।

(A) $5 \pm \sqrt{6}$ (B) $3\sqrt{3}$ (C) 5 (D) $\sqrt{5} \pm 6$

Ans. **A**

Sol. A, B = A + d, C = A + 2d

$\Rightarrow 3A + 3d = 180^\circ \quad \Rightarrow A + d = 60^\circ = B$

$$\cos 60^\circ = \frac{a^2 + 100 - 81}{20a} = \frac{1}{2}$$

$\Rightarrow a^2 + 19 = 10a \quad \Rightarrow a^2 - 10a + 19 = 0$

$\Rightarrow a^2 - 10a + 25 = 6 \quad \Rightarrow (a-5)^2 = 6 \quad \Rightarrow a = 5 \pm \sqrt{6}\,(A)$

3. In ΔABC, the line joining the circumcentre to the incentre is parallel to BC, then cos B + cos C =

त्रिभुज ABC में परिकेन्द्र तथा अन्तःकेन्द्र को जोड़ने वाला रेखाखण्ड BC के समान्तर है, तो cos B + cos C का मान होगा।

(A) 3/2 (B)1 (C) 3/4 (D) 1/2

Ans. **B**

Sol. $\Rightarrow R\cos A = r \Rightarrow \cos A = \frac{r}{R} = -\cos(B+C)$

$$\cos B + \cos C = \frac{c^2 + a^2 - b^2}{2ca} + \frac{a^2 + b^2 - c^2}{2ab}$$

$$\Rightarrow R\left(1 - 2\sin^2\frac{A}{2}\right) = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$$

$\Rightarrow \cos A = \cos A + \cos B + \cos C - 1$

$\Rightarrow \cos B + \cos C = 1$ (B)

4. The perimeter of a ΔABC is 6 times the arithmetic mean of the sines of its angles. If the a = 1, then the angle A is

त्रिभुज ABC में इसका परिमाप कोणों की ज्या (sine) के समान्तर माध्य का 6 गुना है। यदि a = 1 है, तो कोण A का मान होगा।

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) π

Ans. **A**

Sol. $2s = 6 \times \left(\frac{\sin A + \sin B + \sin C}{3}\right) \Rightarrow \frac{a+b+c}{2} = \left(\frac{a+b+c}{2R}\right)$

$\Rightarrow R = 1$

$\Rightarrow a = 2R\sin A \Rightarrow 1 = 2\sin A \Rightarrow A = 30^\circ$ (A)

5. If a = 15 , b = 36 and c = 39 , then the value of cos A is equal to :

(A) $\frac{12}{13}$ (B) $\frac{5}{13}$ (C) $\frac{4}{13}$ (D) None of these

यदि a = 15 , b = 36 तथा c = 39 तो cos A का मान होगा।

(A) $\frac{12}{13}$ (B) $\frac{5}{13}$ (C) $\frac{4}{13}$ (D) इनमें से कोई नहीं

Ans. **A**

Sol. $\cos A = \dfrac{36^2 + 39^2 - 15^2}{2 \times 36 \times 39} = \dfrac{36}{39} = \dfrac{12}{13}(A)$

6. If D is the mid point of the side BC of a triangle ABC and AD is perpendicular to AC, then :
यदि D भुजा BC का मध्य बिन्दु है तथा AD, AC पर लम्ब है तो त्रिभुज ABC के लिए निम्न में से कौनसा सत्य है।
(A) $3b^2 = a^2 - c^2$ (B) $3a^2 = b^2 - 3c^2$ (C) $b^2 = a^2 - c^2$ (D) $a^2 + b^2 = 5c^2$

Ans. **A**

Sol. $\Rightarrow \cos \angle CAD = \dfrac{b^2 + d^2 - \dfrac{a^2}{4}}{2bd} = 0$ & $d = \dfrac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$

$\Rightarrow b^2 + dd^2 = \dfrac{a^2}{4} \Rightarrow 4b^2 + \dfrac{1}{4}\left(2b^2 + 2c^2 - a^2\right) = \dfrac{a^2}{4}$

$\Rightarrow 6b^2 + 2c^2 - 2a^2 = 0 \Rightarrow 3b^2 = a^2 - c^2$ (A)

7. There exists a triangle ABC satisfying

(A) $\tan A + \tan B + \tan C = 0$ (B) $\dfrac{\sin A}{2} = \dfrac{\sin B}{3} = \dfrac{\sin C}{7}$

(C) $(a + b)^2 = c^2 + ab$ (D) None

यदि एक त्रिभुज ABC विद्यमान है, तो निम्न में से कौनसा सत्य हैं।

(A) $\tan A + \tan B + \tan C = 0$ (B) $\dfrac{\sin A}{2} = \dfrac{\sin B}{3} = \dfrac{\sin C}{7}$

(C) $(a + b)^2 = c^2 + ab$ (D) कोई नहीं

Ans. **C**

Sol. $a^2 + b^2 + ab = c^2$ $\Rightarrow \dfrac{a^2 + b^2 - c^2}{2ab} = -\dfrac{1}{2} = \cos C$ (C)

$\tan (B + C) - (\tan B + \tan C) = 0$ $\Rightarrow (\tan B + \tan C)\left(\dfrac{1}{1 - \tan B \tan C} - 1\right) = 0$

$\Rightarrow \tan B + \tan C = 0$ or $\tan B \tan C = 0$

8. If in a triangle ABC angles A, B, C are in A.P. and $2b^2 = 3c^2$, then least angle will be :
यदि त्रिभुज ABC में कोण A, B, C स.श्रे. में है तथा $2b^2 = 3c^2$ तो त्रिभुज का सबसे छोटा कोण होगा :
(A) 15º (B) 30º (C) 45º (D) 50º

Ans. **C**

Sol. A, B, C $\rightarrow$ AP $\Rightarrow B = 60^\circ$ & $2b^2 = 3c^2$

$\cos B = \dfrac{c^2 + a^2 - b^2}{2ca} = \dfrac{1}{2}$ $\Rightarrow c^2 + a^2 - \dfrac{3c^2}{2} = ca$

$\Rightarrow 2a^2 - c^2 = 2ca \Rightarrow 2\left(\dfrac{a}{c}\right)^2 - 2\left(\dfrac{a}{c}\right) - 1 = 0$

$\dfrac{a}{c} = \dfrac{2 \pm \sqrt{4 + 8}}{4} = \dfrac{1 \pm \sqrt{3}}{2}$

$$\Rightarrow \frac{a}{c}=\frac{\sqrt{3}+1}{2}=\frac{\sin A}{\sin C}$$

$$\Rightarrow \text{smallest side} \Rightarrow c \Rightarrow \cos c=\frac{a^2+b^2-c^2}{2ab}$$

$$\Rightarrow \frac{\sin(C+2d)}{\sin C}=\frac{\sqrt{3}+1}{2}$$

$$\text{Now, } \frac{b^2}{\sin^2 B}=\frac{c^2}{\sin^2 C}\Rightarrow \frac{2b^2}{2\sin^2 B}=\frac{3c^2}{3\sin^2 C}$$

$$\Rightarrow \sin^2 C=\frac{2}{3}\times\left(\frac{3}{4}\right)=\frac{1}{2}\Rightarrow \sin C=\frac{1}{\sqrt{2}}\Rightarrow c=45^\circ \text{ (C)}$$

9. If radius of the circumcircle of a Δ ABC is 4 cm and D, E, F are the feet of perpendiculars drawn from the vertices to the opposite sides, then radius of the circumcircle of Δ DEF is :

(A) 3 cm (B) 1 cm (C) 2 cm (D) None of these

यदि Δ ABC के परिवृत्त की त्रिज्या 4 cm है तथा विपरीत भुजाओं पर शीर्षों से डाले गये लम्ब पाद D, E, F है तो Δ DEF के परिवृत्त की त्रिज्या होगी।

(A) 3 cm (B) 1 cm (C) 2 cm (D) कोई नहीं

Ans. **C**

Sol. $R=4$ cm, $\Rightarrow R'=\frac{R}{2}=\frac{4}{2}=2$ cm. (C)

10. In a triangle ABC if $\frac{r}{r_1}=\frac{1}{2}$, then the value of $\tan\frac{A}{2}\left(\tan\frac{B}{2}+\tan\frac{C}{2}\right)$ is equal to :

(A) 2 (B) 1/2 (C) 1 (D) None of these

त्रिभुज ABC में यदि $\frac{r}{r_1}=\frac{1}{2}$ तो $\tan\frac{A}{2}\left(\tan\frac{B}{2}+\tan\frac{C}{2}\right)$ बराबर होगा –

(A) 2 (B) 1/2 (C) 1 (D) कोई नहीं

Ans. **B**

Sol. $$\frac{r}{r_1}=\frac{1}{2}\Rightarrow \frac{\frac{\Delta}{s}}{\frac{\Delta}{s-a}}=\frac{1}{2}\Rightarrow \frac{s-a}{s}=\frac{1}{2}\Rightarrow 1-\frac{a}{s}=\frac{1}{2}\Rightarrow \frac{a}{s}=\frac{1}{2}\Rightarrow a=\frac{s}{2}$$

$$\Rightarrow \tan\frac{A}{2}\left(\tan\frac{B}{2}+\tan\frac{C}{2}\right)=\frac{\Delta}{s(s-a)}\left[\frac{\Delta}{s(s-b)}+\frac{\Delta}{s(s-c)}\right]=\frac{\Delta^2}{s^2(s-a)(s-b)(s-c)}[s-c+s-b]$$

$$=\frac{1}{s}[2s-(b+c)]=\frac{a}{s}=\frac{1}{2}(B)$$

11. If in triangle ABC, $(a+b+c)(a+b-c)=\lambda ab$, then exhaustive range of 'λ' is :

त्रिभुज ABC में यदि $(a+b+c)(a+b-c)=\lambda ab$ तो 'λ' का परिसर होगा :

(A) (2 , 4) (B) (0 , 4) (C) (0 , 2) (D) (1 , 4)

Ans. **B**

Sol. $\left(a^2+b^2+2ba-c^2\right)=\lambda ab$

$a^2 + b^2 - c^2 = \lambda ab - 2ab = ab(\lambda - 2)$

$$\frac{a^2+b^2-c^2}{ab}=\lambda-2$$

$$\frac{a^2+b^2-c^2}{2ab}=\frac{\lambda-2}{2}$$

$$\cos C=\frac{\lambda-2}{2}$$

$$-1\le\frac{\lambda-2}{2}\le 1 \qquad \Rightarrow 0\le\lambda\le 4$$

12. In a triangle ABC, if $\frac{a^3+b^3+c^3}{\sin^3 A+\sin^3 B+\sin^3 C}=343$, then diameter of circle circumscribing the triangle is

(A) 14 units (B) 7 units (C) 21 units (D) None of these

त्रिभुज ABC में यदि $\frac{a^3+b^3+c^3}{\sin^3 A+\sin^3 B+\sin^3 C}=343$ तो त्रिभुज के परिवृत्त का व्यास होगा–

(A) 14 इकाई (B) 7 इकाई (C) 21 इकाई (D) कोई नही

Ans. B

Sol. Given $\frac{a^3+b^3+c^3}{\sin^3 A+\sin^3 B+\sin^3 C}=343$

using sine rule

$$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}=2R \rightarrow \text{circumradius}$$

$a = 2R \sin A$

$b = 2R \sin B$

$c = 2R \sin C$

substitute a, b, c in given equation

$$8R^3\left(\frac{\sin^3 A+\sin^3 B+\sin^3 C}{\sin^3 A+\sin^3 B+\sin^3 C}\right)=343$$

$(2R)^3 = 7^3$

$2R = 7$

diameter of circle = 7 unit

13. If a, b and c be in A.P. prove that cos A $\cot\frac{A}{2}$, cos B $\cot\frac{B}{2}$, and cos C $\cot\frac{C}{2}$ are in A.P.

यदि a, b तथा c A.P. में है, तो सिद्ध कीजिए cos A $\cot\frac{A}{2}$, cos B $\cot\frac{B}{2}$, तथा cos C $\cot\frac{C}{2}$ भी A.P. में होगें।

Ans.

Sol. $\cos A\cdot\cot\frac{A}{2}=\left(1-2\sin^2\frac{A}{2}\right)\frac{\cos\frac{A}{2}}{\sin\frac{A}{2}}=\cot\frac{A}{2}-2\sin\frac{A}{2}\cos\frac{A}{2}=\cot\frac{A}{2}-\sin A$

$\cot\frac{A}{2}=\frac{s(s-a)}{\Delta}\Rightarrow\cot\frac{A}{2},\cot\frac{B}{2},\cot\frac{C}{2}\rightarrow AP$

$\Rightarrow\frac{s(s-a)}{\Delta},\frac{s(s-b)}{\Delta},\frac{s(s-c)}{\Delta}\rightarrow AP$

$\Rightarrow s-a,\ s-b,\ s-c\rightarrow AP$

$\Rightarrow +a,+b,+c\rightarrow AP$

$\Rightarrow \sin A,\sin B,\sin C\rightarrow AP$

$\Rightarrow\frac{a}{2R},\frac{b}{2R},\frac{c}{2R}\rightarrow AP\Rightarrow a,b,c\rightarrow AP$

14. If a, b and c are in H.P. prove that $\sin^2\frac{A}{2},\sin^2\frac{B}{2}$ and $\sin^2\frac{C}{2}$ are also in H.P.

यदि a, b तथा c H.P. में है, तो सिद्ध कीजिए $\sin^2\frac{A}{2},\sin^2\frac{B}{2}$ तथा $\sin^2\frac{C}{2}$ भी H.P. में होगे।

Ans.

Sol. $\sin^2\frac{A}{2}=\frac{(s-b)(s-c)}{bc}\Rightarrow\sin^2\frac{A}{2},\sin^2\frac{B}{2},\sin^2\frac{C}{2}\rightarrow HP$

$\Rightarrow\sin^2\frac{B}{2}=\frac{2\sin^2\frac{A}{2}\cdot\sin^2\frac{C}{2}}{\sin^2\frac{A}{2}+\sin^2\frac{C}{2}}\Rightarrow\frac{(s-c)(s-a)}{ca}=\frac{2\times\frac{(s-b)(s-c)(s-a)(s-b)}{bc\times ab}}{\frac{(s-b)(s-c)}{bc}+\frac{(s-a)(s-b)}{ab}}$

$\Rightarrow\frac{(s-b)}{b}\left[\frac{(s-c)}{c}+\frac{(s-a)}{a}\right]=\frac{2(s-b)^2}{b^2}\Rightarrow\frac{s}{c}-1+\frac{s}{a}-1=\frac{2s}{b}-2$

$\Rightarrow\frac{1}{a}+\frac{1}{c}=\frac{2}{b}\Rightarrow a,b,c\rightarrow HP$

15. Two sides of a triangle are given by the roots of the equation , $x^2-2\sqrt{3}\,x+2=0$. The angle between the sides is $\frac{\pi}{3}$. Find the perimeter of the triangle .

त्रिभुज की दो भुजाऐं समीकरण $x^2-2\sqrt{3}\,x+2=0$ के मूलों द्वारा निरूपित की जाती है तथा इनके बीच का कोण $\frac{\pi}{3}$ है, तो त्रिभुज का परिमाप होगा।

Ans. $2\sqrt{3}+\sqrt{6}$

Sol. $x^2 - 2\sqrt{3}x + 3 = 1 \Rightarrow \left(x - \sqrt{3}\right)^2 = 1 \Rightarrow x - \sqrt{3} = \pm 1 \Rightarrow x = \sqrt{3} \pm 1$

$\Rightarrow \cos A = \dfrac{\left(\sqrt{3}+1\right)^2 + \left(\sqrt{3}-1\right)^2 - a^2}{2\left(3-1\right)} = \dfrac{1}{2}$ Þ $a^2 = 8 - 2 = 6 \Rightarrow a = \sqrt{6}$

$\Rightarrow 2s = a + b + c = \sqrt{3} + 1 + \sqrt{3} - 1 + \sqrt{6} = 2\sqrt{3} + \sqrt{6}$

1. If in a triangle ABC angle B = 90°, then $\tan^2 \frac{A}{2}$ is

(A) $\frac{b-c}{a}$ (B) $\frac{b-c}{b+c}$ (C) $\frac{b+c}{b-c}$ (D) None of these

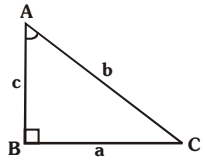
यदि त्रिभुज ABC में कोण B = 90° है तो $\tan^2 \frac{A}{2}$ बराबर होगा –

(A) $\frac{b-c}{a}$ (B) $\frac{b-c}{b+c}$ (C) $\frac{b+c}{b-c}$ (D) कोई नहीं

Ans. **B**

Sol. $\tan^2 \frac{A}{2} = \frac{1-\cos A}{1+\cos A}$

$= \frac{1-\frac{c}{b}}{1+\frac{c}{b}} = \frac{b-c}{b+c}$



2. In a triangle ABC , CH and CM are the lengths of the altitude and median to the base AB .
If a = 10 , b = 26 , c = 32 then length (HM) is :
(A) 5 (B) 7 (C) 9 (D) None of these
त्रिभुज ABC , CH तथा CM, भुजा AB पर शीर्षलम्ब तथा माध्यिका है।
यदि a = 10 , b = 26 , c = 32 तो (HM) की लम्बाई होगी :
(A) 5 (B) 7 (C) 9 (D) कोई नहीं

Ans. **C**

Sol. BM = 16

$\cos B = \frac{10^2+32^2-26^2}{2\times10\times32} = \frac{100+6\times58}{640} = \frac{448}{640} = \frac{7}{10}$

$\Rightarrow BH = BC\cos B = 10\times\frac{7}{10} = 7$

$\Rightarrow$ HM = BM – BH = 16 – 7 = 9 (C)

3. The product of the arithmetic mean of the lengths of the sides of a triangle and harmonic mean of the lengths of the altitudes of the triangle is equal to :
त्रिभुज की भुजाओं के समान्तर माध्य तथा शीर्ष लम्बों का ह.माध्य का गुणनफल बराबर होगा :
(A) Δ (B) 2Δ (C) 3Δ (D) 4Δ

Ans. **B**

Sol. AD = c sin B = b sin C
BE = a sin C = c sin A
CF = b sin A = a sin B

$$\Rightarrow \left(\frac{a+b+c}{3}\right)\left(\frac{3}{\frac{1}{a\sin B}+\frac{1}{b\sin C}+\frac{1}{c\sin A}}\right)=\frac{(a+b+c)}{2R\left(\frac{1}{ab}+\frac{1}{bc}+\frac{1}{ca}\right)}$$

$$=\frac{(a+b+c)}{2\times\left(\frac{abc}{4\Delta}\right)}\times\frac{abc}{(a+b+c)}=\frac{4\Delta}{2}=2\Delta \quad \text{(B)}$$

4. In a ΔABC a = 5 , b = 4 and c = 3 . 'G' is the centroid of ΔABC. Circumradius of ΔGAB is equal to :
ΔABC a = 5 , b = 4 तथा c = 3 यदि 'G', ΔABC का केन्द्रक है तो ΔGAB की परित्रिज्या होगी :

(A) $2\sqrt{13}$ (B) $\frac{5}{12}\sqrt{13}$ (C) $\frac{5}{3}\sqrt{13}$ (D) $\frac{3}{2}\sqrt{13}$

Ans. **B**

Sol. $AM=\frac{1}{2}\sqrt{18+32-25}$

$$=\frac{1}{2}\sqrt{25}=\frac{5}{2}$$

$$AG=\frac{2}{3}\times\frac{5}{2}=\frac{5}{3}$$

Now, $\tan\theta=\frac{2}{3}\Rightarrow\sin\theta=\frac{2}{\sqrt{13}}\Rightarrow R_{GAB}=\frac{AG}{2\sin\theta}=\frac{5}{3}\times\frac{\sqrt{13}}{2\times2}$

$$\Rightarrow R_{GAB}=\frac{5\sqrt{13}}{12} \qquad \text{(B)}$$

5. In a triangle ABC , if $r_1 = 2r_2 = 3r_3$, then show that $\frac{a}{b}=\frac{5}{4}$.

यदि त्रिभुज ABC में यदि $r_1 = 2r_2 = 3r_3$ तो सिद्ध कीजिये कि $\frac{a}{b}=\frac{5}{4}$.

Ans.

Sol. $r_1 = 2r_2 = 3r_3 \qquad \Rightarrow PT: \frac{G}{b}=\frac{5}{4}$

$$\Rightarrow r_1=\frac{\Delta}{(s-a)}=\frac{2\Delta}{(s-b)}=\frac{3\Delta}{(s-c)}$$

$\Rightarrow s - b = 2s - 2a$
$2s - 2c = 3s - 3b$
$\Rightarrow s = 3b - 2c$
$\Rightarrow a + b + c = 6b - 4c$
$\Rightarrow 5c = 5b - a$
$\Rightarrow 15a - 15b = 5b - a$

$\Rightarrow 16a = 20b \Rightarrow \frac{a}{b} = \frac{5}{4}$

$s - c = 3s - 3a \Rightarrow c = 3a - 2s$
$\Rightarrow s = 3b - 6a + 4s \Rightarrow 3s = 6a - 3b \Rightarrow s = 7a - b$
$\Rightarrow a + b + c = 4a - 2b \Rightarrow c = 3a - 3b$

6. If I is the incentre of ΔABC, then show that the ratio $IA : IB : IC = \text{cosec}\ \frac{A}{2} : \text{cosec}\ \frac{B}{2} : \text{cosec}\ \frac{C}{2}$

यदि I, ΔABC का अन्तःकेन्द्र हो तो सिद्ध कीजिये कि अनुपात $IA : IB : IC = \text{cosec}\ \frac{A}{2} : \text{cosec}\ \frac{B}{2} : \text{cosec}\ \frac{C}{2}$

Ans.

Sol. $IA = r\,\text{cosec}\frac{A}{2}$

$\Rightarrow IA : IB : IC = r\,\text{cosec}\frac{A}{2} : r\,\text{cosec}\frac{B}{2} : r\,\text{cosec}\frac{C}{2}$

7. If g, h, k denotes the sides of a pedal triangle, prove that : $\frac{g}{a^2} + \frac{h}{b^2} + \frac{k}{c^2} = \frac{a^2 + b^2 + c^2}{2abc}$

यदि g, h, k त्रिभुज ABC के पदिक त्रिभुज की भुजाऐं है तो सिद्ध कीजिये : $\frac{g}{a^2} + \frac{h}{b^2} + \frac{k}{c^2} = \frac{a^2 + b^2 + c^2}{2abc}$

Ans.

Sol. $\Rightarrow a = 2R \sin A \quad , \quad a' = g = R \sin 2A$

$$\Rightarrow LHS = \frac{g}{a^2} + \frac{h}{b^2} + \frac{R}{c^2} = \frac{R\sin 2A}{4R^2\sin^2 A} + \frac{R\sin 2B}{4R^2\sin^2 B} + \frac{R\sin 2C}{4R^2\sin^2 C}$$

$$= \frac{2R\sin A\cos A}{2hR^2\sin^2 A} + \frac{2R\sin B\cos B}{2hR^2\sin^2 B} + \frac{2R\sin c\cos C}{2hR^2\sin^2 C} = \frac{\cot A + \cot B + \cot C}{2R}$$

$$= \frac{\frac{b^2 + c^2 - a^2}{4\Delta} + \frac{c^2 + a^2 - b^2}{4\Delta} + \frac{a^2 + b^2 - c^2}{4\Delta}}{2 \times \left(\frac{abc}{4\Delta}\right)} = \frac{a^2 + b^2 + c^2}{2abc} = RHS$$

8. Two regular polygons of ' n ' and ' 2 n ' sides have the same perimeter, show that their areas are in the ratio, $2\cos\frac{\pi}{n} : 1+\cos\frac{\pi}{n}$.

दो समबहुभुज जिनकी भुजाऐं ' n ' तथा ' 2 n ' है के समान परिमाप है तो सिद्ध कीजिये उनके क्षेत्रफल का अनुपात $2\cos\frac{\pi}{n} : 1+\cos\frac{\pi}{n}$ है।

Ans.

Sol. (n) $\theta = \frac{\pi}{n} \Rightarrow \tan\frac{\pi}{n} = \frac{\frac{a}{2}}{H} \Rightarrow H = \frac{a}{2}\cot\frac{\pi}{n}$

$$\Rightarrow A_1 = n\times\frac{1}{2}\times a\times\frac{a}{2}\omega t\left(\frac{\pi}{n}\right) = \frac{na^2}{4}\omega t\left(\frac{\pi}{n}\right)$$

(2n) $\theta = \frac{\pi}{2n} \Rightarrow A_2 = \frac{nb^2}{2}\omega t\left(\frac{\pi}{2n}\right)$

$$\Rightarrow \frac{A_1}{A_2} = \frac{a^2\cot\left(\frac{\pi}{n}\right)}{2b^2\omega t\left(\frac{\pi}{2n}\right)} = \frac{2\cos\left(\frac{\pi}{n}\right)}{2\sin\left(\frac{\pi}{2n}\right)\cos\left(\frac{\pi}{2n}\right)}\times\frac{\sin\left(\frac{\pi}{2n}\right)}{\cos\left(\frac{\pi}{2n}\right)}$$

$$= \frac{\cos\left(\frac{\pi}{n}\right)}{\cos^2\left(\frac{\pi}{2n}\right)} = \frac{\cos\left(\frac{\pi}{n}\right)}{\frac{1+\cos\left(\frac{\pi}{n}\right)}{2}}$$

$$\Rightarrow A_1 : A_2 = 2\cos\left(\frac{\pi}{n}\right) : \left(1+\cos\frac{\pi}{n}\right)$$

9. If 2a be the side of a regular polygon of ' n ' sides, R and r be the circumradius and inradius, prove that $R + r = a\cot\frac{\pi}{2n}$.

यदि 2a किसी ' n ' भुजाओं वाले समबहुभुज की एक भुजा है। यदि R तथा r परिवृत्त तथा अन्तःवृत्त की त्रिज्याऐं हो तो सिद्ध कीजिये $R + r = a\cot\frac{\pi}{2n}$.

Ans.

Sol. $\theta = \frac{\pi}{n} \qquad \Rightarrow R = a \operatorname{cosec}\theta \qquad ; \qquad r = a\cot\theta$

$\Rightarrow R + r = a(\cot\theta + \operatorname{cosec}\theta)$

$$\Rightarrow R + r = a\left(\frac{1+\cos\theta}{\sin\theta}\right) = a \times \frac{2\cos^2\left(\frac{\theta}{2}\right)}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}} = a\cot\frac{\theta}{2} = a\cot\left(\frac{\pi}{2n}\right) = \text{RHS}$$

Prove that (Que. 10 to 16)

सिद्ध कीजिये (Que. 10 to 16)

10. $\frac{b-c}{r_1} + \frac{c-a}{r_2} + \frac{a-b}{r_3} = 0$

Ans.

Sol. $$\frac{b-c}{r_1} + \frac{c-a}{r_2} + \frac{a-b}{r_3} = \frac{(b-c)(s-a)}{\Delta} + \frac{(c-a)(s-b)}{\Delta} + \frac{(a-b)\times(s-c)}{\Delta}$$

$$= \frac{sb - sc - ab + ca + sc - sa - bc + ab + sa - sb - ca + bc}{\Delta} = 0$$

11. $a\cot A + b\cot B + c\cot C = 2(R + r)$

Ans.

Sol. $$a\cot A + b\cot B + c\cot C = 2R\sin A\frac{\cos A}{\sin A} + 2R\cos B + 2R\cos C$$

$$= 2R(\cos A + \cos B + \cos C) = 2R\left(1 + \frac{r}{R}\right) = 2(R + r)$$

12. $\frac{r_1}{(s-b)(s-c)} + \frac{r_2}{(s-c)(s-a)} + \frac{r_3}{(s-a)(s-b)} = \frac{3}{r}$

Ans.

Sol. $$\frac{r_1}{(s-b)(s-c)} + \frac{r_2}{(s-c)(s-a)} + \frac{r_3}{(s-a)(s-b)} = \frac{\Delta s}{s(s-a)(s-b)(s-c)} + \frac{\Delta s}{\Delta^2} + \frac{s}{\Delta} = \frac{3}{\left(\frac{\Delta}{s}\right)} = \frac{3}{r}$$

13. $\frac{r_1 - r}{a} + \frac{r_2 - r}{b} = \frac{c}{r_3}$

Ans.

Sol. $\dfrac{r_1-r}{a}+\dfrac{r_2-r}{b}=\underbrace{\dfrac{\Delta}{s-a}-\dfrac{\Delta}{s}}_{a}+\underbrace{\dfrac{\Delta}{s-b}-\dfrac{\Delta}{s}}_{b}=\Delta\left[\dfrac{a}{s\cdot a(s-a)}+\dfrac{b}{s\cdot b(s-b)}\right]=\dfrac{\Delta}{s(s-a)}+\dfrac{\Delta}{s(s-b)}$

$=\dfrac{\Delta}{s}\left[\dfrac{2s-(a+b)}{(s-a)(s-b)}\right]=\dfrac{\Delta c\times(s-c)}{s(s-a)(s-b)(s-c)}=\dfrac{c(s-c)}{\Delta}=\dfrac{c}{\left(\dfrac{\Delta}{(s-c)}\right)}=\dfrac{c}{r_3}$

14. $\left(\dfrac{1}{r}-\dfrac{1}{r_1}\right)\left(\dfrac{1}{r}-\dfrac{1}{r_2}\right)\left(\dfrac{1}{r}-\dfrac{1}{r_3}\right)=\dfrac{4R}{r^2s^2}$

Ans.

Sol. $\dfrac{(r_1-r)(r_2-r)(r_3-r)}{r^3r_1r_2r_3}=\dfrac{(r_1r_2r_3)-(r_1r_2+r_2r_3+r_3r_1)r+(r_1+r_2+r_3)r^2-r^3}{r^3r_1r_2r_3}$

$=\dfrac{\left[\dfrac{\Delta^2}{\Delta s}\right]-\left[s^2\times\dfrac{\Delta}{\Delta s}\right]+(4R+r)r^2-r^3}{r^2\times\Delta^2}=\dfrac{4Rr^2+r^3-r^3}{r^2\Delta^2}=\dfrac{4R}{\Delta^2}=\dfrac{4R}{r^2s^2}$

15. $\left(\dfrac{1}{r}+\dfrac{1}{r_1}+\dfrac{1}{r_2}+\dfrac{1}{r_3}\right)^2=\dfrac{4}{r}\left(\dfrac{1}{r_1}+\dfrac{1}{r_2}+\dfrac{1}{r_3}\right)$

Ans.

Sol. $\left(\dfrac{1}{r}+\dfrac{1}{r_1}+\dfrac{1}{r_2}+\dfrac{1}{r_3}\right)^2=\left(\dfrac{1}{r}+\dfrac{1}{r}\right)^2=\left(\dfrac{2}{r}\right)^2=\dfrac{4}{r^2}=\dfrac{4}{r}\times\dfrac{1}{r}=\dfrac{4}{r}\left(\dfrac{1}{r_1}+\dfrac{1}{r_2}+\dfrac{1}{r_3}\right)$

16. $\cot\dfrac{A}{2}+\cot\dfrac{B}{2}+\cot\dfrac{C}{2}=\dfrac{s^2}{\Delta}$

Ans.

Sol. $\cot\dfrac{A}{2}+\cot\dfrac{B}{2}+\cot\dfrac{C}{2}=\dfrac{s(s-a)}{\Delta}+\dfrac{s(s-b)}{\Delta}+\dfrac{s(s-c)}{\Delta}=\dfrac{s}{\Delta}[3s-2s]=\dfrac{s^2}{\Delta}$

17. Given a ΔABC with sides $a = 7,\ b = 8$ and $c = 5$. If the value of the expression $\left(\sum\sin A\right)\left(\sum\cot\dfrac{A}{2}\right)$ can be expressed in the form $\dfrac{p}{q}$ (lowest form) where $p, q \in N$ find the value of $(p + q)$.

एक त्रिभुज ABC में भुजाऐं $a = 7,\ b = 8$ तथा $c = 5$ है। यदि व्यंजक $\left(\sum\sin A\right)\left(\sum\cot\dfrac{A}{2}\right)$ का मान $\dfrac{p}{q}$ के न्यूनतम समकक्ष रूप में लिखा गया है तो $(p + q)$ का मान क्या होगा, जहाँ $p, q \in N$?

Ans. 107

Sol. $\Delta ABC \Rightarrow \sum \sin A = \frac{a+b+c}{2R} = \frac{2s}{2R} = \frac{s}{R}$ & $\sum \cot \frac{A}{2} = \frac{s^2}{\Delta}$

$\Rightarrow \left(\sum \sin A\right)\left(\sum \cot \frac{A}{2}\right) = \frac{s}{R} \times \frac{s^2}{\Delta} = \frac{s^3}{\Delta} \times \frac{4\Delta}{(abc)} = \frac{4 \times (1000)}{5 \times 7 \times 8} = \frac{100}{7}$

18. In a triangle ABC, if the sides a, b, c are roots of $x^3 - 11x^2 + 38x - 40 = 0$. If $\sum\left(\frac{\cos A}{a}\right) = \frac{p}{q}$, then find the least value of $(p + q)$.

एक त्रिभुज ABC में, यदि भुजायें a, b, c समीकरण $x^3 - 11x^2 + 38x - 40 = 0$ के मूल है। यदि $\sum\left(\frac{\cos A}{a}\right) = \frac{p}{q}$ है, तो $(p + q)$ का न्यूनतम मान ज्ञात कीजिये।

Ans. 25

Sol. $x^3 - 11x^2 + 38x - 40 = 0 \rightarrow a, b, c$

$\Rightarrow \sum\left(\frac{\cos A}{a}\right) = \frac{\left(b^2 + c^2 - a^2\right) + \left(c^2 + a^2 - b^2\right) + \left(a^2 + b^2 - c^2\right)}{2abc} = \frac{a^2 + b^2 + c^2}{2abc}$

$= \frac{(a+b+c)^2 - 2(ab + bc + ca)}{2abc}$

$= \frac{(11)^2 - 2(38)}{2 \times (40)} = \frac{121 - 76}{80} = \frac{45}{80} = \frac{9}{16}$